

2402-COL719

Project: 3D-memory(TSV)
Milestone 2

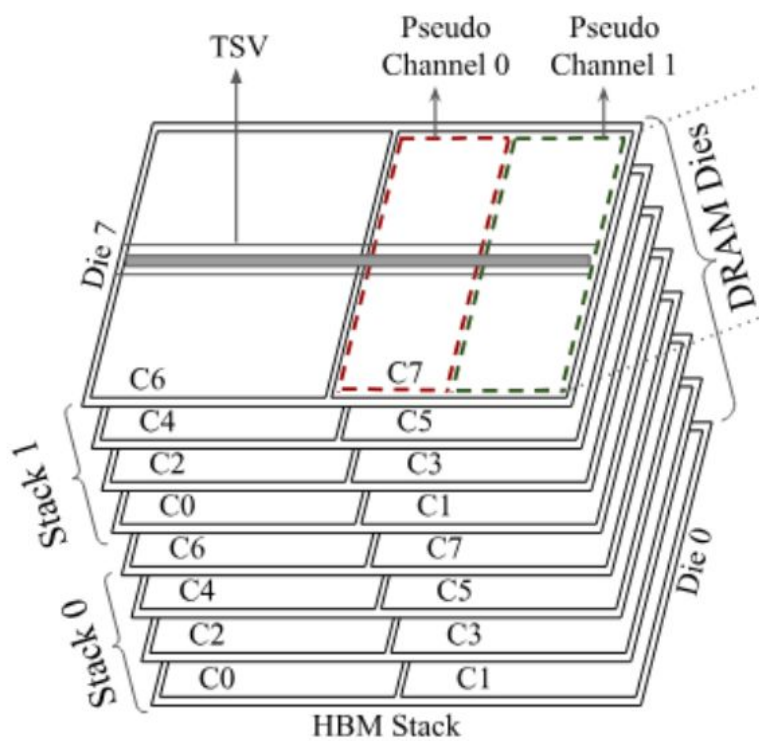
Submission of

- Mohd Sayanur Rahman
(2021PH10221)
- Osho Anand (2023EET2176)

Objective

- Understand HBM working
- Experiment with different synthetic workloads utilizing 16 cores and 8 channels HBM
- Design a suitable Dynamic Thermal Management(DTM) policy.

High Bandwidth Memory(HBM) with Through Silicon Via(TSV)



- 4 X 4 X 8 banks
- 16 banks per channel
- Horizontally placed banks/ranks form channel.
- Channels at the bottom 4 layers are duplicated at the top 4 layers
- 8 channels (16 Pseudochannels)
- Each channel is independent of other.
- Various memory interleaving policies to map physical address to banks/ranks in and across channels
- Various access scheduling policies and channel partitioning policies

Simulation description

- Simulation interval : 1ms
- Timestep = simulation interval /1000 = 1us
- Temperature gradient is solved using Hotspot tool, which is provided a power trace file, layer configuration file that includes floorplan files for each layer, an initial temperature file and a configuration file for heatsink and leakage power model.
- Grid points are the centers of each banks.
- Power trace is calculated using model involving constant power per read/write access and constant refresh power.
- The experiment involve trying different read-write access rates sequence, that approximates various testbenches.

Workloads description

- Random sequence of numbers as rates of memory access trace where :
 - Memory-intensive program have access rates in range 10,000/s to 11,000/s.
 - Memory-lightweight programs have most access rates in range 0 to 2,000/s while some(5%) in 7,000/s to 10,000/s.
 - Mixed intensity programs have half of the numbers in 0 to 4,000/s and rest in 4,000/s to 8,000/s throughout simulation.
- Each core is allocated a separate Pseudo-channels of HBM, to reduce memory interference. ¹
- The Workloads comprises:
 - Memory-intensive programs running on all cores throughout the simulation.
 - Memory-intensive programs running on 8 cores, and mixed-intensity on other 8 cores.
 - Memory-intensive on 4 cores, Mixed-intensity on 8 cores, Light-intensity on 4 cores.
 - Mixed-intensity on 8 cores and Light-Intensity on 8 cores.
 - Light-intensity on all (16 cores)

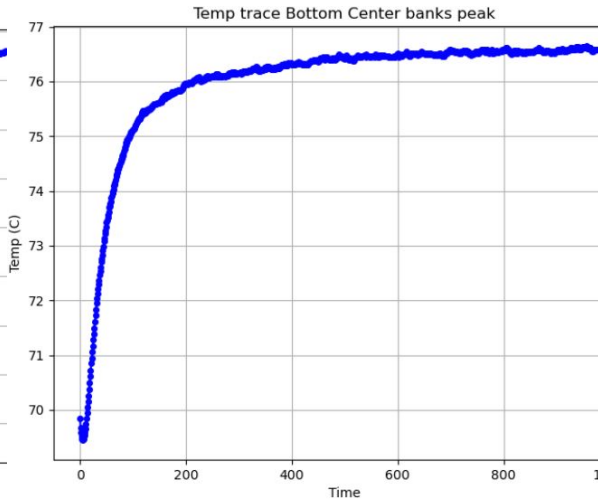
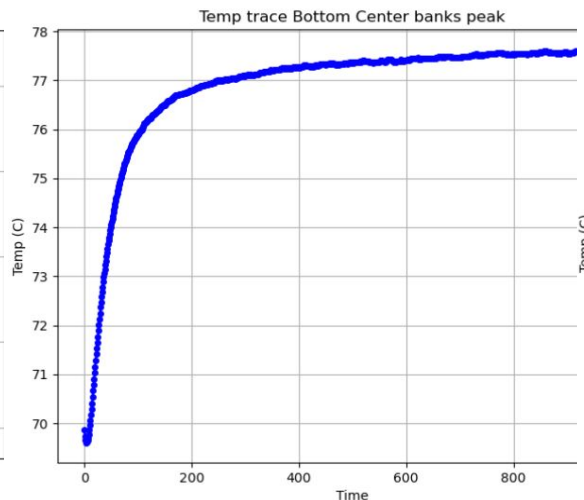
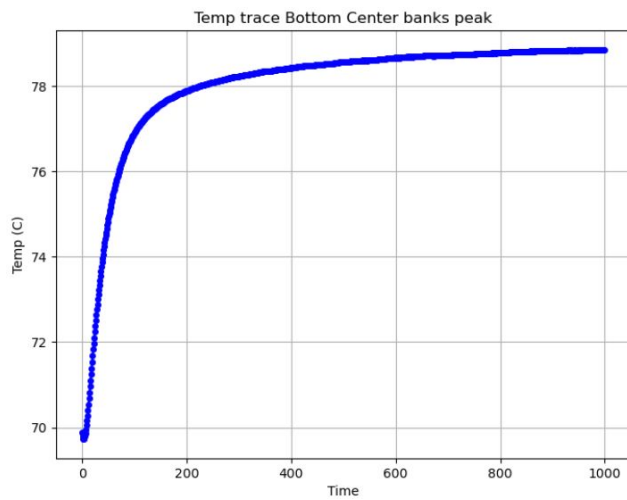
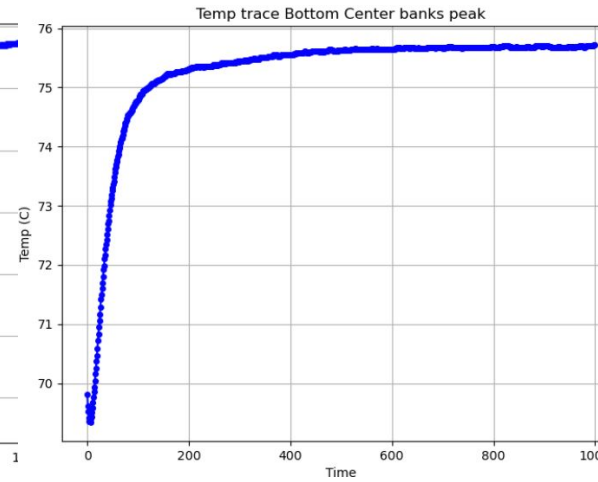
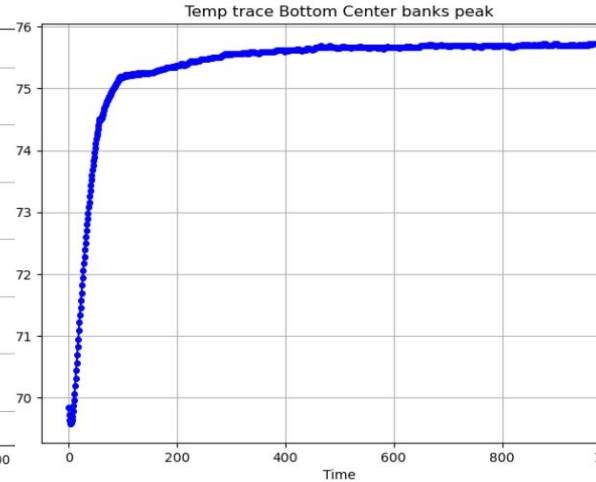
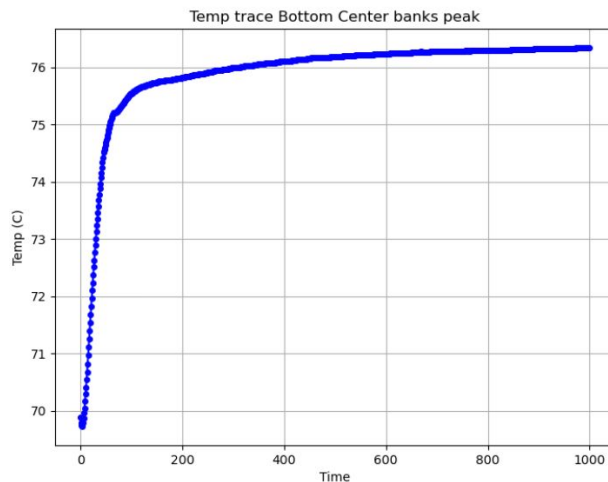
Dynamic Thermal Management (DTM) Policy

- A primitive DTM is tested to be effective enough to just keep the peak temperatures below thresholds.
- The Policy solely involves putting Hot-channels (i.e. peak temp. of banks of the channel exceeds threshold) to Low-Power mode(LPM) that involves preventing accesses (and thus dynamic power dissipation).
- Policy description:
 - Channels with peak temp. exceeding T_0 is put to LPM. If the channel cools down to $T_{0_c}(<T_0)$, it is put in active state.
 - If channel's peak temp. further exceeds $T_1(>T_0)$, another neighbouring channel in contact with the highest temp. is put to LPM and brought back to active state once peak temp. goes below $T_{1_c}(<T_1 \text{ and } >T_0)$.
 - Another neighbouring channel is put to LPM if the channel's peak temp further exceeds $T_2(>T_1)$ and brought back to active state once peak temp. goes below $T_{2_c}(<T_2 \text{ and } >T_1)$.
 - This continues until all 4 closest neighbours are put to LPM beyond T_4 .
 - $T_{0_c} < T_0 < T_{1_c} < T_1 < T_{2_c} < T_2 < T_{3_c} < T_3 < T_{4_c} < T_4$
 - The threshold values are required to be optimized on the testbench.
- Drawback: Half of the cores are stalled most of the time and the policy is not application-aware.

With
DTM

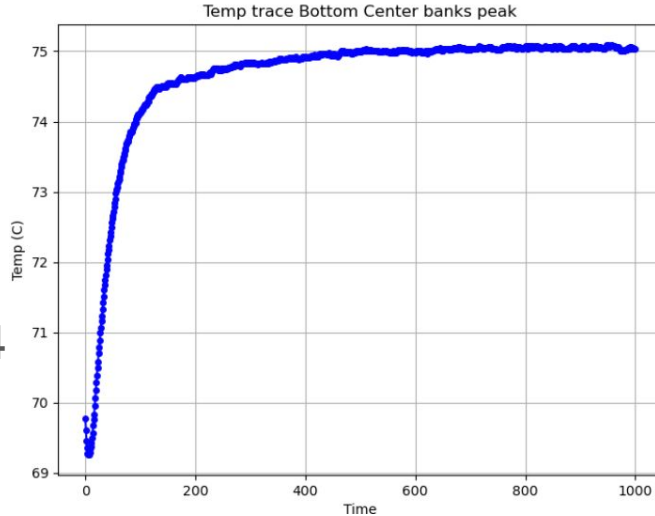
Bottom
face
center
banks

No
DTM



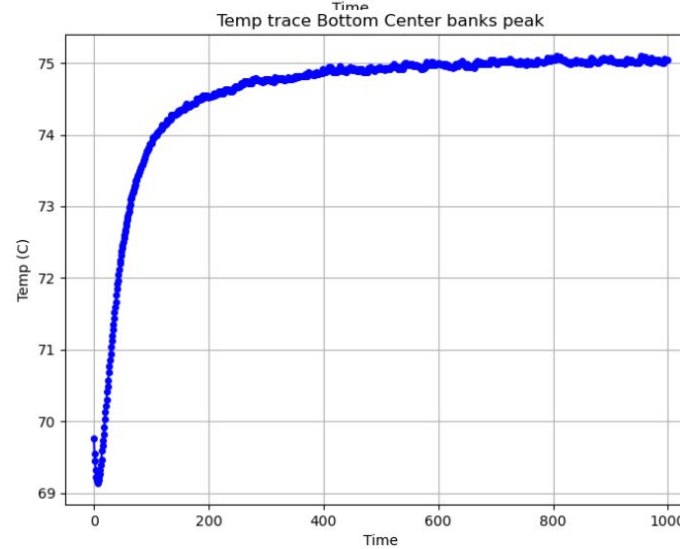
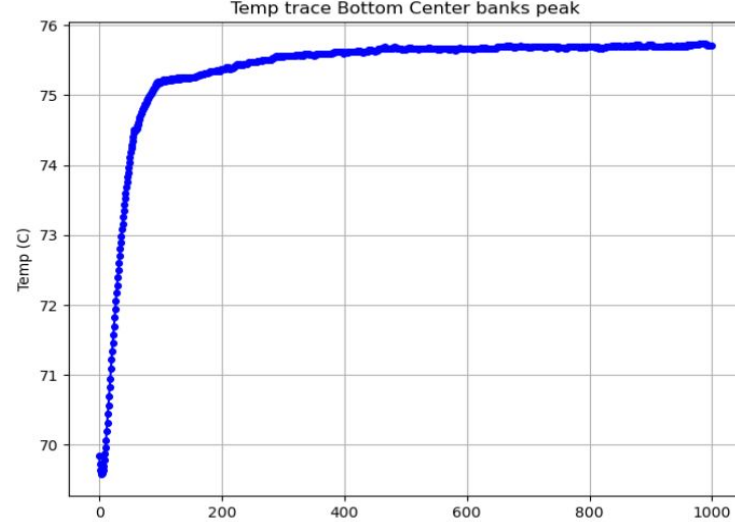
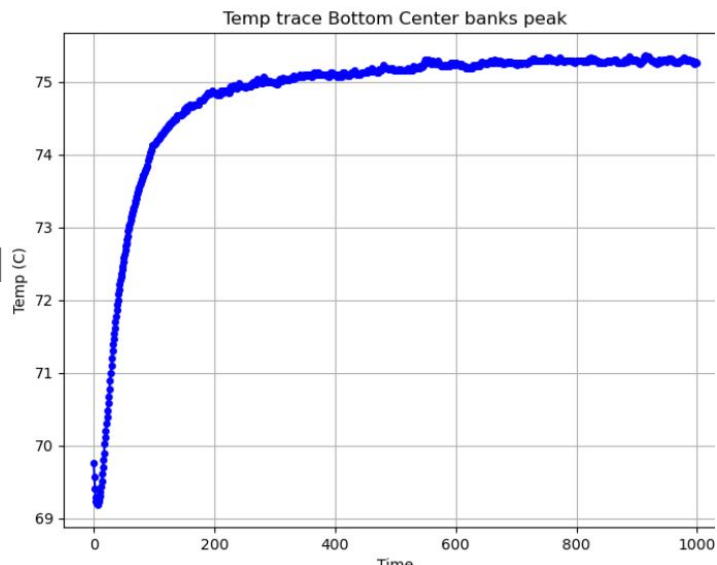
With
DTM

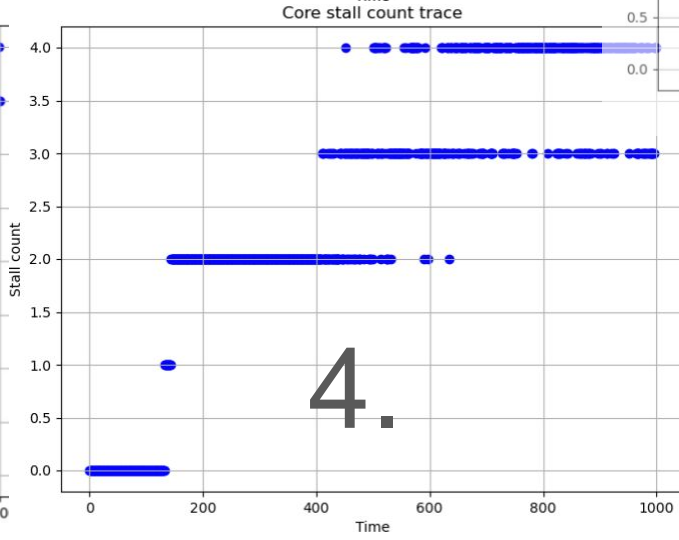
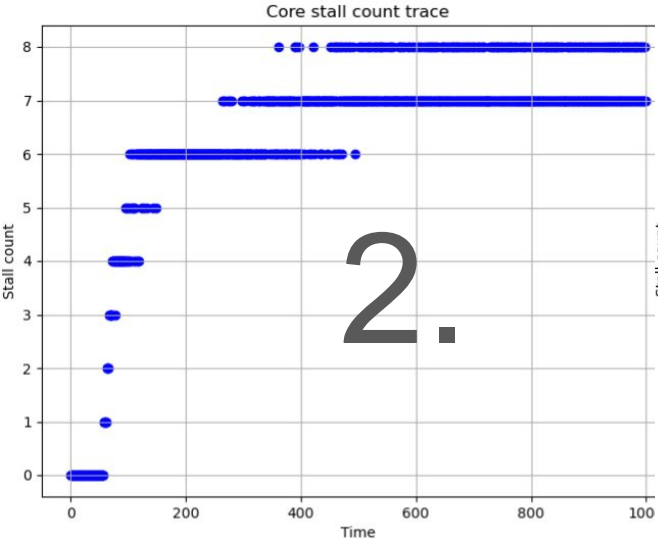
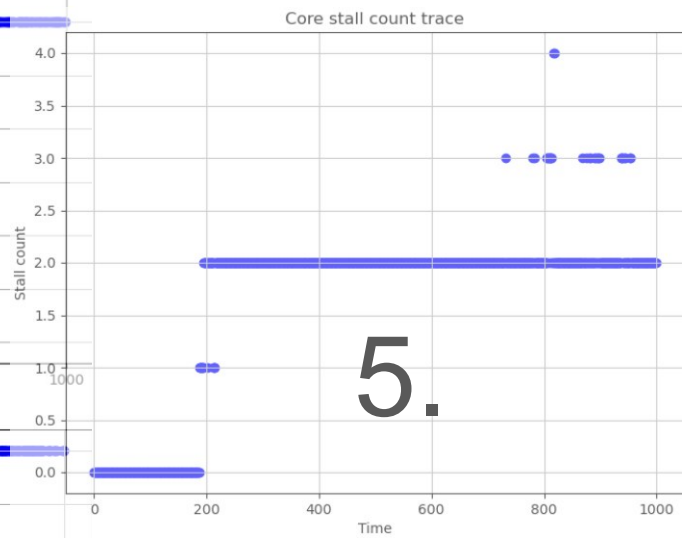
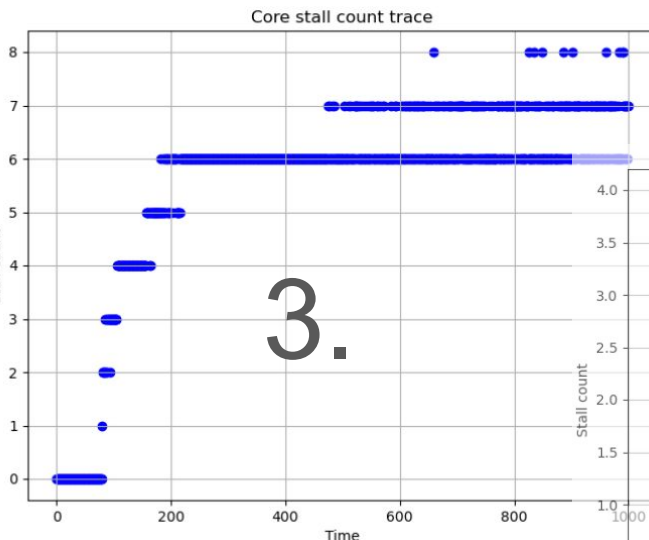
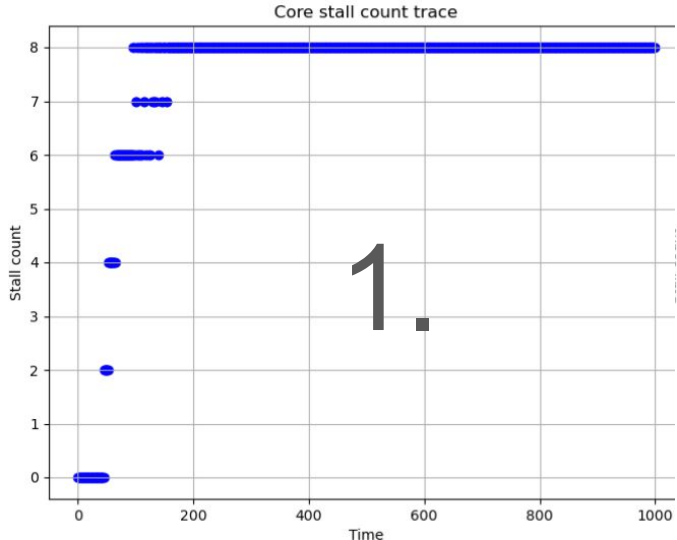
Workload 4



No
DTM

Workload 5





Stalled cores
over time-
Each plot is of
a workload

Observations

- The implemented DTM policy is keeping the memory's peak temp a bit less than the case without DTM for given workloads, at the cost of stalling nearly half of the cores (for memory-intense workload) for most of the time.

Further Work

- More experiments with different DTMs to get the best result. Current DTM is much primitive and is not showing any good result.
- We plan to include data migration strategies as well, so we need to model the overhead caused due to migration.
- We might need to include some more workloads, because current testbench is not analyzed yet if it covers most workload instances.
- For better DTM policies, we need to first identify the critical channels or banks, and thus need to come up with some application-aware DTM policy.

For all of the plots, visit here :

https://github.com/heysayan/COL719_assignment_comet/tree/proj719/hotspot_tool/visualization